18. Two straight lines $\vec{r}=\vec{a}+s\vec{b}$ and $\vec{r}=\vec{c}+t\vec{d}$ intersect each other if $(\vec{c}-\vec{a})\cdot(\vec{b}\times\vec{d})=0$

19. The shortest distance between the two parallel lines $\vec{r}=\vec{a}+s\vec{b}$ and $\vec{r}=\vec{c}+t\vec{b}$ is $d=\dfrac{|(\vec{c}-\vec{a})\times\vec{b}|}{|\vec{b}|}$, where $|\vec{b}|\neq 0$

20. If two lines $\dfrac{x-x_1}{b_1}=\dfrac{y-y_1}{b_2}=\dfrac{z-z_1}{b_3}$ and $\dfrac{x-x_2}{d_1}=\dfrac{y-y_2}{d_2}=\dfrac{z-z_2}{d_3}$ intersect, then

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix}=0$$

21. A straight line which is perpendicular to a plane is called a normal to the plane.

22. The equation of the plane at a distance p from the origin and perpendicular to the unit normal vector $\hat{d}$ is $\vec{r}\cdot\hat{d}=p$ (normal form)

23. Cartesian equation of the plane in normal form is $lx+my+nz=p$

24. Vector form of the equation of a plane passing through a point with position vector $\vec{a}$ and perpendicular to $\vec{n}$ is $(\vec{r}-\vec{a})\cdot\vec{n}=0$.

25. Cartesian equation of a plane normal to a vector with direction ratios a,b,c and passing through a given point (x_1,y_1,z_1) is $a(x-x_1)+b(y-y_1)+c(z-z_1)=0$.

26. Intercept form of the equation of the plane $\vec{r}\cdot\vec{n}=q$, having intercepts a,b,c on the x,y,z axes respectively is $\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}=1$.

27. Parametric form of vector equation of the plane passing through three given non-collinear points is $\vec{r}=\vec{a}+s(\vec{b}-\vec{a})+t(\vec{c}-\vec{a})$

28. Cartesian equation of the plane passing through three non-collinear points is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix}=0.$$

29. A straight will lie on a plane if every point on the line, lie in the plane and the normal to the plane is perpendicular to the line.

30. The two given non-parallel lines $\vec{r}=\vec{a}+s\vec{b}$ and $\vec{r}=\vec{c}+t\vec{d}$ are coplanar if $(\vec{c}-\vec{a})\cdot(\vec{b}\times\vec{d})=0$.

31. Two lines $\dfrac{x-x_1}{b_1}=\dfrac{y-y_1}{b_2}=\dfrac{z-z_1}{b_3}$ and $\dfrac{x-x_2}{d_1}=\dfrac{y-y_2}{d_2}=\dfrac{z-z_2}{d_3}$ are coplanar if

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix}=0$$

32. Non-parametric form of vector equation of the plane containing the two coplanar lines $\vec{r}=\vec{a}+s\vec{b}$ and $\vec{r}=\vec{c}+t\vec{d}$ is $(\vec{r}-\vec{a})\cdot(\vec{b}\times\vec{d})=0$ or $(\vec{r}-\vec{c})\cdot(\vec{b}\times\vec{d})=0$.

33. The acute angle θ between the two planes $\vec{r}\cdot\vec{n}_1=p_1$ and $\vec{r}\cdot\vec{n}_2=p_2$ is $\theta=\cos^{-1}\left(\dfrac{|\vec{n}_1\cdot\vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}\right)$

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